

What do forecasters say they don't know?

Elicited macroeconomic uncertainty in the US and euro-area Surveys of Professional Forecasters, 1968–2026 — working draft

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Abstract

Professional forecasters state their own uncertainty four times a year. Since 1968 the US Survey of Professional Forecasters has collected probability distributions — not just point forecasts — for output growth and inflation, and the ECB's survey has done the same for the euro area since 1999. We pool every reported histogram by the law of total variance, decomposing total uncertainty into the average forecaster's stated uncertainty and the disagreement between forecasters, for every variable, horizon, and survey round — 3,695 round-target distributions. Five findings organize the panorama. Stated growth uncertainty is stable: its pandemic spike decayed by 2026 on both continents, while euro-area inflation uncertainty remains a third above its pre-2021 level. Disagreement — the only uncertainty visible in point-forecast data — is the minority share everywhere: a sixth of total variance for growth, at most a third for any variable. Stated uncertainty stops rising with horizon by year two, and the ten-year US point forecasts invert the term structure outright: forecasters put a tighter interquartile range on the next decade's growth (0.2pp) than on next year's (0.4pp). The US distributions are roughly calibrated on average (69.7% one-sigma coverage over 33 years); the euro area's are overconfident; both fail in clusters at regime shifts, in the tail the regime moved toward — and scored properly, the densities beat climatology everywhere but beat a consensus-plus-historical-error Gaussian only for growth and the euro area. Density elicitation stops three to five years out. Published estimates of AI's growth effects span roughly 0.1 to 30 percentage points a year (Shenk 2026; Cunningham 2025) at the horizons no survey asks about, and we show what asking would take. A live interactive tracks every series.

1 Introduction

In the first quarter of 2026, the forecasters in the US Survey of Professional Forecasters put a median of 1.8% on next year's real GDP growth, with an interquartile range of 0.4 points across their stated distributions. Asked about average growth over the next ten years, the same panel's answers had a median of 2.1% and an interquartile range of 0.2 points. The decade answer is tighter than the one-year answer — and no survey asked the decade question as a distribution. The ten-year numbers are point forecasts; the histograms stop three years out.

That gap matters more now than at any point in the surveys’ history. Published estimates of what artificial intelligence will do to growth over roughly the coming decade run from under 0.1 percentage points a year (Acemoglu 2024) through 1.5 points (Briggs and Kodnani 2023) — with a 0.07–1.24-point spread inside a single framework (Aghion and Bunel 2024) — to above 30 points (Davidson 2021; Erdil and Besiroglu 2023). Compilations put the full published range at roughly 0.1–30 points a year, almost entirely as point estimates (Shenk 2026; Cunningham 2025). The one large elicitation, run by the Forecasting Research Institute with the Federal Reserve Bank of Chicago, collected medians and two quantiles, once (Karger et al. 2026). The instruments that would discipline this debate — surveys that make professionals state distributions and then score them — exist, run quarterly, and have five decades of track record. They stop at year three in the US and year five in the euro area.

This paper assembles that track record in full. We compute one construction — the law-of-total-variance decomposition of pooled survey histograms (Wallis 2005; Boero et al. 2008) — for every density variable, every horizon, and every round of the two long-running professional surveys, 3,695 round-target distributions in all, and evaluate every one against realized outcomes where a verified realization concept exists (3,070 forecast-outcome pairs). The distinction we track back to 1968 is Zarnowitz and Lambros’s (1987): *disagreement* — the spread between forecasters’ means — is not *uncertainty* — the spread each forecaster reports around their own mean.

i Draft note

Contribution list and positioning paragraph drafted in the planning skeleton ([paper/PAPER.md](#)); nearest antecedents: D’Amico and Orphanides (2008) (US long series, inflation, ends 2008), Clements et al. (2022) (both surveys, in prose), Allayioti et al. (2024) (ECB 25-year retrospective, one survey).

2 The elicitation landscape

Two surveys regularly ask professional forecasters for probability distributions over macroeconomic outcomes. The US SPF has collected histograms since 1968Q4 — for nominal GNP growth at first, real GNP from 1981Q3, real GDP from 1992Q1 — for output growth and the GDP price index throughout, joined by fourth-quarter-over-fourth-quarter core CPI and core PCE inflation in 2007Q1 and the unemployment rate in 2009Q2 (Federal Reserve Bank of Philadelphia 2026). Density horizons ran to the next calendar year until 2009Q2 and to three years out since. The ten-year variables — real GDP growth, CPI, PCE — are point forecasts only. The ECB SPF has run quarterly since 1999Q1: densities for HICP inflation, real GDP growth, and unemployment (core HICP since 2017Q1) at current-year through year-after-next calendar targets, rolling one- and two-year horizons, and a longer-term calendar year four years ahead in Q1/Q2 rounds and five ahead in Q3/Q4 (García 2003; Allayioti et al. 2024).

The rest of the survey landscape collects points. The NY Fed’s consumer survey elicits household densities to three years; Consensus Economics elicits by-range probabilities at one year and publishes its 5–10-year forecasts as points; Livingston (since 1946) and Blue Chip are point

products; the FOMC’s projections come with fan charts built from historical errors, not elicited beliefs. The two long-horizon precedents are one-off quantile elicitations (Christensen et al. 2018; Karger et al. 2026). No survey anywhere regularly elicits probability distributions over ten-year-ahead growth.

3 Measures

Each forecaster attaches probabilities to bins. We place each bin’s mass at its midpoint, give open tails the width of the adjacent bin, keep responses whose probabilities sum to within two points of 100, and renormalize. For forecaster i , the histogram yields a mean m_i and a variance v_i . The law of total variance splits the pooled mixture (Wallis 2005):

$$\text{total variance} = \mathbb{E}[v_i] + \text{Var}(m_i).$$

The first term is the average uncertainty forecasters state individually; the second is disagreement, the only term visible in point-forecast data (Zarnowitz and Lambros 1987; Lahiri and Sheng 2010). We report the square roots, the between share, and quantiles of the pooled mixture. The interquartile range of the mixture doubles as a robustness measure: bin-scheme changes move midpoint-based SDs mechanically (Shoja and Soofi 2017; Knüppel and Pavlova 2026), and the 2020Q2 US bin widening is the worked example.

Targets are calendar years, so a “next year” forecast made in Q1 and one made in Q4 differ in horizon by three quarters (Ganics et al. 2024). We treat round-target pairs explicitly and show the within-year shrinkage — current-year uncertainty falls 24–41% from Q1 to Q4, depending on the variable — rather than averaging over it (Clements et al. 2025).

Coverage rates test the density at two thresholds; proper scores evaluate all of it (Gneiting and Raftery 2007; Boero et al. 2011). The pooled CDF is piecewise linear, so the continuous ranked probability score has an exact closed form, and pinball losses at the published quantile levels decompose it by tail. We score every calibrated round against two no-lookahead benchmarks — expanding-window climatology and a Gaussian centered on the consensus with historical-error RMSE — and against the average individual forecaster. One structural note: the pooled CDF is the mean of individual CDFs and the CRPS is convex in the CDF, so the pool beats the average individual on every observation by construction; the empirical content is the size of that gain (9.2% on average, 3.5–18.7% by cell) and where it thins.

4 Five stylized facts

i Draft note

Each fact gets a figure from the pipeline. Inline values below compute from `outputs/` at render; the balanced-sample term-structure and post-2021 ECB comparisons cite the pipeline reports (`SOL_REPORT.md`, `SCORES_REPORT.md`) pending their move into computed chunks.

Stated growth uncertainty is stable — and its pandemic spike decayed on both continents, while euro-area inflation uncertainty has not come home. The pooled SD of next-year US

growth was 1.24 points in 1992 and averaged 1.26 through 2020; it peaked at 2.10 in 2021 and stands at 1.36 in 2026 (decade means: 1.25 over 2010–19, 1.74 over 2020–24). Euro-area growth uncertainty spiked higher and decayed as fast. Euro-area inflation uncertainty did not: the one-year series remains a third above its pre-2021 level in 2026.

Disagreement is the minority share everywhere, but not uniformly. The median between share is about a sixth for growth (US 17%, euro area 16%) and euro-area prices (14%), and a quarter to a third for unemployment and the US inflation densities (core CPI 32%, core PCE 32%). The spread visible in point forecasts understates stated uncertainty two-to-threefold across the whole panorama (Rich and Tracy 2010; Giordani and Söderlind 2003).

Stated uncertainty rises with horizon only until year two – then plateaus, and for growth it inverts. US growth within-SD climbs from the current year to two years out, then gains a point in the hundredths place at three. In the euro area, total growth uncertainty is mildly higher at the one-year rolling horizon than at the four-to-five-year horizon. Only unemployment keeps steepening. And the ten-year US point forecasts invert the term structure outright: IQR 0.2 for the decade against 0.4 for next year in 2026.

The US distributions are roughly calibrated on average; the euro area’s are overconfident; both fail in tails, in clusters. A $\pm 1\sigma$ band around the consensus covered 23 of 33 next-year US growth outcomes (69.7%) over 1993–2025 – textbook – but the misses cluster: six straight target years around the late-1990s productivity boom, then 2008–09, then 2020–21. Euro-area one-sigma coverage runs 49–57% by variable (Kenny et al. 2014; 2015). Scored properly, the densities beat climatology everywhere (US growth +40% mean CRPS skill) but beat the consensus-plus-track-record Gaussian only for US growth (+6%) and the euro area; for US inflation and unemployment, the elicited shape adds nothing beyond the point forecast wrapped in its own error history (Clements 2018; Knüppel and Krüger 2022). The pinball decomposition puts the 2008–09 growth losses in the lower tail and the 2020–22 inflation losses in the upper tail by factors of five to nine.

The euro area asks the question the US doesn’t – and after 2021 the long answer moved less than the short one, but it moved. The ECB longer-term density’s total SD rose about 31% after 2021, against 59% at one year; its consensus moved from 1.8% to 2.0% while the one-year consensus moved from 1.6% to 2.3% (Górnicka and Meyler 2022; Vincent-Humphreys et al. 2019). Stated long-run uncertainty now sits at its series high exactly where the US elicits nothing.

5 The elicitation gap

i Draft note

Section drafted at beat level. The argument: put the verified 0.1–30pp/yr published range (Shenk 2026; Cunningham 2025) and its anchors (Acemoglu 2024; Briggs and Kodnani 2023; Aghion and Bunel 2024; Davidson 2021; Erdil et al. 2025; Korinek and Suh 2024) against the inventory of section 2; Karger et al. (2026) as the one-off flagship; Chow et al. (2026) extracting long-horizon growth beliefs from asset prices precisely because no survey elicits them. The observation the paper owns: the instruments that would discipline this debate exist, run quarterly, have 50 years of calibration record — and stop at year three (US) or five (euro area).

6 What better elicitation would look like

i Draft note

Design beats, not advocacy: extend one existing survey block to a ten-year growth density (the ECB’s longer-term question shows a panel will answer it, and its 2023 special survey asked whether ten-year expectations would differ (Allayioti et al. 2024)); bins wide enough to be honest about tails (the 2020 widening as precedent (European Central Bank 2020)); ship the question with the scoring rule that disciplines it — CRPS with pinball components — because the record shows stated densities break at regime shifts and the AI question is a regime-shift question; pair each long-horizon density with the respondent’s AI-adoption assumption so the 0.1–30 range becomes decomposable.

7 Conclusion

i Draft note

One page: the 2026 pair again — 0.4pp next year, 0.2pp for the decade — with the whole panorama behind it.

Data and limitations

All measures compute from the public microdata of the Philadelphia Fed SPF (Federal Reserve Bank of Philadelphia 2026) and the ECB SPF, with bin schemes taken from each survey’s documentation for every era. Realizations are latest-vintage (Stark 2010); real-time-vintage evaluation is future work. Forecasts of a shared outcome are correlated, so miss counts and score comparisons are descriptive. Benchmark windows are expanding with a ten-observation minimum; euro-area annual histories are short. The pipeline, tests, and every series behind the live interactive are in the paper’s repository.

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